

Damien MAÎTRE

MSc Engineering Student — CFD, HPC & Numerical Methods

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PROFILE

MSc engineering student specializing in fluid dynamics, structural mechanics and high performance computing, with hands-on experience across CFD, FEM and infrastructure engineering. Seeking an engineering internship in the energy industry, tackling physical problems, grid management or project finance, starting March 2027.

EDUCATION

September 2026 February 2027	MSc Physics engineering <i>Student exchange at Politecnico di Milano, Italy</i> International economics, Energy engineering, High performance computing, Infrastructure finance, Neural Networks.
September 2023 June 2027	MSc Engineering — Fluid dynamics, Structural mechanics, Prototyping <i>École nationale des ponts et chaussées (ENPC), Paris</i> Structural mechanics and large-structure design, FEM applied to structural analysis, Composite materials, Metallurgy, Complete fluid dynamics module, HPC, Combustion. • Tools: OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, gmsh • Projects: Underwater sea-glider simulation and prototype ; closed-wing aircraft design and optimization ; wooden suspension bridge design, verification and construction.

RESEARCH PROJECTS

Complete CFD developer-first suite

Personal project — OpenFOAM, Gmsh (forked), Python, C++

Open and user-friendly aerodynamical loads studies, tailored for very high Re axisymmetric problems.

Structural FEM & Eurocode Verification suite

Personal project — OpenSeesPy, Python

End to end structure design, visualization and verification tools.

PROFESSIONAL EXPERIENCE

February 2026 July 2026	Infrastructure engineering internship — TRACTEBEL FRANCE <i>CDG Express project, Saint-Ouen</i> • Construction compliance verification for the CDG Express, a 2.5B€ high-speed rail project, • Structural analysis and Eurocode verifications of civil engineering structures, • Review of structural calculation reports and as-built site surveys, • Drafting technical reports and recommendations for the client.
July 2025 January 2026	Data & Industry Project Manager — SNCF Voyageurs <i>Industry & digital solutions, Saint-Denis</i> • Management of a 150k€ digital project within the rolling stock maintenance industry, • Coordination of 15 3D artists across the country's maintenance plants, • Project ownership and reporting to internal clients (plant directors, Industrial Director, etc.), • End-to-end procurement and tendering process.

TECHNICAL SKILLS

CFD & FEM Programming	OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, Gmsh 10+ years of experience with Python, C++ and UNIX environments
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LANGUAGES

French — Native · **English** — Proficient (C1) · **Italian** — Working proficiency (B2) · **German** — Working proficiency (B2)

ACTIVITIES & COMMUNITY INVOLVEMENT

Rescuer — French Red Cross

Volunteer first-aid responder

First-aid responder at public events (marathons, festivals, Paris-Plage) and on-call with Paris Fire Brigade.

Director — Student Musical Ensemble

BridgeWay Comédie Musicale, Champs-Sur-Marne

Directed an ensemble of 75 artists and a creative/technical leadership team of 12 with 4.5k€ budget.

General Secretary — Student Association

Union des Elèves de l'Ecole Nationale des Ponts et Chaussées

Management of a team of 15 for professional and social events with 1,000+ attendees, Administrative management of a registered French non-profit (Loi 1901).